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Studierichting: Informatica
Oefeningenreeks 5A nr. 11.1

$$\int_0^{\frac{\pi}{2}} x^2 \sin x dx$$

$$\left\{\begin{array}{l} u = x^2 \\ dv = \sin x dx \end{array}\right. \Rightarrow \left\{\begin{array}{l} du = 2xdx \\ v = -\cos x \end{array}\right.$$

$$\stackrel{PI}{=} [-x^2 \cos x]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x \cdot 2xdx$$

$$= \left(-\frac{\pi}{4} \cdot 0\right) + 2 \int_0^{\frac{\pi}{2}} x \cos x dx$$

$$\left\{\begin{array}{l} u = x \\ dv = \cos x dx \end{array}\right. \Rightarrow \left\{\begin{array}{l} du = dx \\ v = \sin x \end{array}\right.$$

$$\stackrel{PI}{=} 2 \cdot \left([x \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x dx \right)$$

$$= 2 \left(\left(\frac{\pi}{2} \cdot 1\right) + [\cos x]_0^{\frac{\pi}{2}} \right) = 2 \left(\frac{\pi}{2} + (0-1)\right)$$

$$= \pi - 2$$

References